

iQ3R Grade 3 Mathematics

*Lesson 3 Module 1 Topic A*¹

*Instructor Version*²

June 10, 2022

This document describes the specific details for the third lesson of Module 1 in our adapted curriculum for 1:1 synchronous instruction in Grade 3 Mathematics³. Topic A is intended to help students develop increasing intuition about the meaning of multiplication. This is the final lesson in topic A.

¹ Adapted for individualized instruction from the EngageNY curriculum for New York State public and private instructors.

² Instructor notes will appear along the side column.

³ This is only an approximate estimate. Students will vary in skill level and preparedness

Student View: Slide 1

Lesson Overview

OBJECTIVE: Interpret the meaning of factors — the size of the group or the number of groups.

AGENDA

- Sprint
- Add to Multiply
- Arrays and Multiplication
- Number Bonds and Factors
- Problem Set
- Debrief

Lesson Overview

Standards

ADDRESSED: 3.OA.1, 3.OA.3

FOCUS: 3.OA.1 Interpret products of whole numbers, e.g., interpret 5×7 as the total number of objects in 5 groups of 7 objects each. For example, describe a context in which a total number of objects can be expressed as 5×7 .

OBJECTIVES: Interpret the meaning of factors—the size of the group or the number of groups. Lesson 3 solidifies students' ability to differentiate the meaning of factors. Students model dividing a whole into equal groups as well as analyze equal groups in scattered configurations and arrays to determine whether factors represent the number of groups or the size of groups. They create pictures, number bonds, and multiplication equations to model their understanding.

SUGGESTED LESSON STRUCTURE

- Fluency Practice — 6 minutes
- Application Problem — 4 minutes
- Concept Development — 20 minutes
- Student Debrief — 10 minutes

Student View: Slide 2

Fluency Practice

Sprint 1

A

Add Equal Groups

1.	$2 + 2 =$	
2.	2 twos =	
3.	$5 + 5 =$	
4.	2 fives =	
5.	$2 + 2 + 2 =$	
6.	3 twos =	
7.	$2 + 2 + 2 + 2 =$	
8.	4 twos =	
9.	$5 + 5 + 5 =$	
10.	3 fives =	
11.	$5 + 5 + 5 + 5 =$	
12.	4 fives =	
13.	2 fours =	
14.	$4 + 4 =$	
15.	2 threes =	
16.	$3 + 3 =$	
17.	2 sixes =	
18.	$6 + 6 =$	
19.	5 twos =	
20.	$2 + 2 + 2 + 2 + 2 =$	

Fluency Practice

Sprint 1

These Sprints are intended to reinforce material learned in Lesson 1. Students will be given 30 seconds to complete as many as possible. Then record the number correct and go on to Sprint 2. Students should be instructed to work as quickly as possible without sacrificing accuracy. They should try to improve on their score in the next sprint.

Student View: Slide 3

Fluency Practice

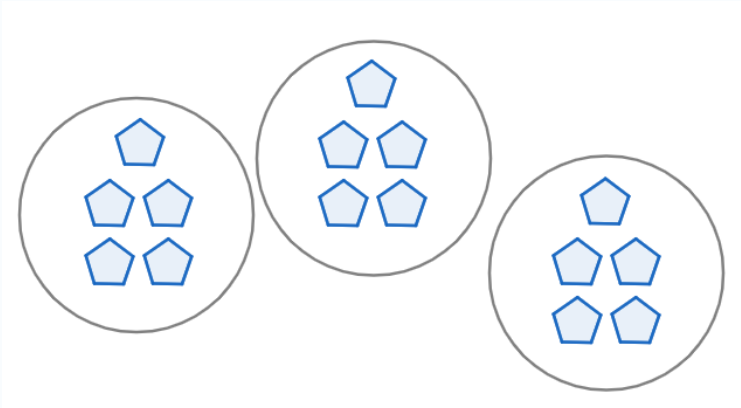
Sprint 2

B		
Add Equal Groups		
1.	5 + 5 =	
2.	2 fives =	
3.	2 + 2 =	
4.	2 twos =	
5.	5 + 5 + 5 =	
6.	3 fives =	
7.	5 + 5 + 5 + 5 =	
8.	4 fives =	
9.	2 + 2 + 2 =	
10.	3 twos =	
11.	2 + 2 + 2 + 2 =	
12.	4 twos =	
13.	2 threes =	
14.	3 + 3 =	
15.	2 sixes =	
16.	6 + 6 =	
17.	2 fours =	
18.	4 + 4 =	
19.	5 fives =	
20.	5 + 5 + 5 + 5 + 5 =	

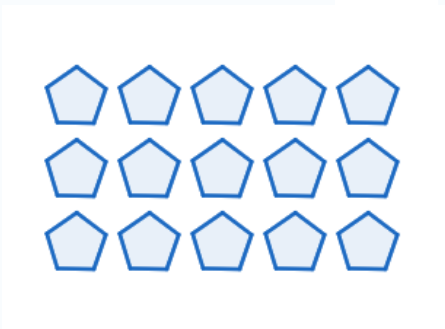
Fluency Practice

Sprint 2

Student View: Slide 4

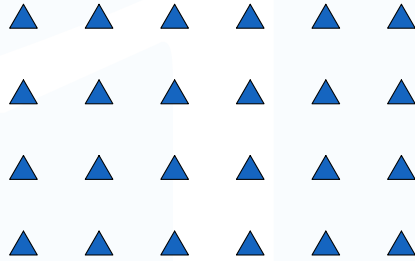
*Fluency Practice**Add to Multiply*

- How many groups are shown in the figure?
- How many items are in each group?
- Write an addition statement that describes the situation.
- Write a multiplication statement representing 3 fives equals 15.
- Can you arrange them in an array?

*Fluency Practice**Add to Multiply*

- This activity can be set up in Geogebra so that students can actually practice arrange them into the array shown below.
- Recall that in an array the “groups” for this picture are the rows.
- The number of things in each row is the number of things in each column. The array should be a 3 row 5 column display as shown in the figure.

Student View: Slide 5

*Concept Development**Arrays and Factors*

IN THE ARRAY above, the number of rows, and the number of triangles in each row are examples of *factors*. The total number of triangles is called the *product* of the factors.

- **Factor** — either the number of equal groups OR number of things in each group.
- **Product** — the total number of things all together.

PRACTICE: Use the array above to help you answer:

1. What factors are represented?
2. What product is represented?
3. What multiplication equation is represented?

*Concept Development**Arrays and Factors*

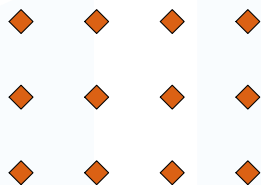
Using the array you will guide the student to the concepts of factors and products. It is very important that they have a concrete representation of the abstract concept of a factor. There are always at least two factors in a multiplication equation. The product is the result. Remember these are all just ways of counting.

- We have selected a fairly large array to illustrate the point that when collections get large, counting things one at a time is not going to work. Eventually we will learn multiplication facts. But these “facts” are simply rules for efficient counting.
- The factors are the number of rows and the number of columns. There are four rows and six columns so the factors are 4 and 6.
- The product is the total number of triangles. There are 24 triangles which can be discerned through a direct count.
- Therefore the array represents the multiplication equation $4 \times 6 = 24$.

Student View: Slide 6

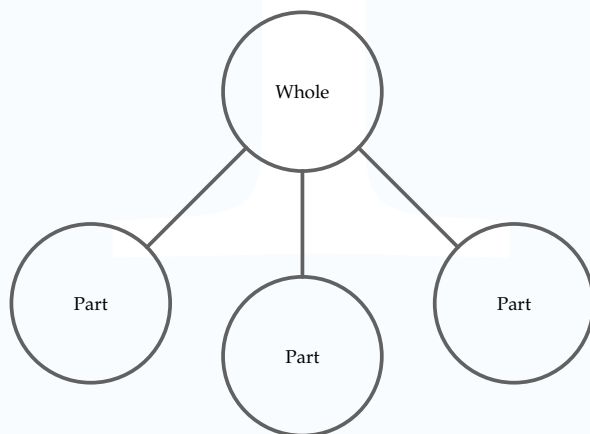
*Concept Development**Number Bonds*

ANOTHER REPRESENTATION you can use for arrays or multiplication problems is called a **number bond**. Consider the array



A number bond “bonds” the whole to its parts.

- whole = 12 red diamonds
- parts = 3 (rows)

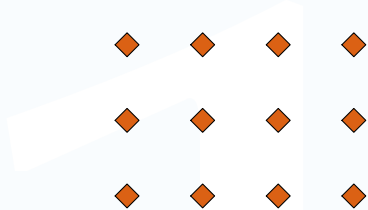
*Concept Development**Number Bonds*

- Number bonds relate a product to its factors. We represent the product as the whole and the number of things in each group as the part.
- The number of groups is represented by the number of circles in the lower part of the diagram. Emphasize the use of the terms “factor” and “product” as you move forward.
- Within each of these diagrams, stress that the numbers in the lower part of the diagram are used to “build” the number on top. In what cases will the number of circles (groups) be greater than the whole?
- In what cases will the numbers in the lower circles be equal to the whole?
- What numbers only allow a bond diagram with three circles?

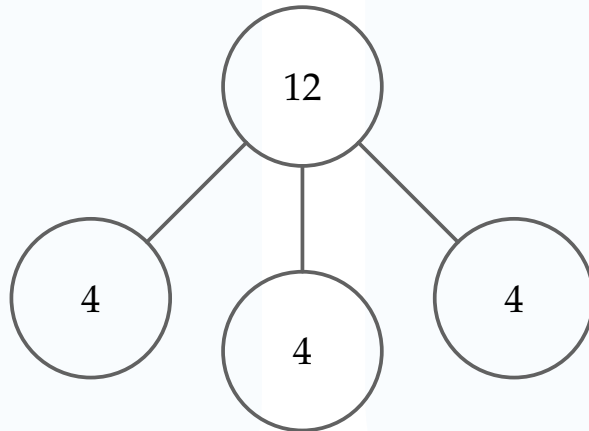
Student View: Slide 7

*Concept Development**Number Bonds*

This array



is represented by this number bond diagram.



- Three fours equal twelve.
- $4 + 4 + 4 = 12$
- $3 \times 4 = 12$
- 3 and 4 are factors of 12.

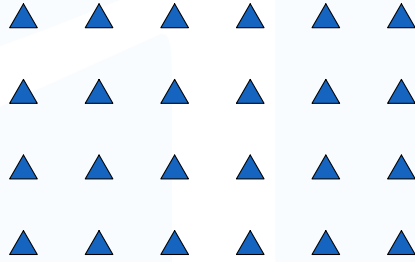
*Concept Development**Number Bonds*

Another good question to ask is how would the number bond diagram differ if we were talking about the multiplication $4 \times 3 = 12$? It would be a different picture, but it would represent the same number. It is important to emphasize this because it gets students ready to understand the commutative property of multiplication from several different representations.

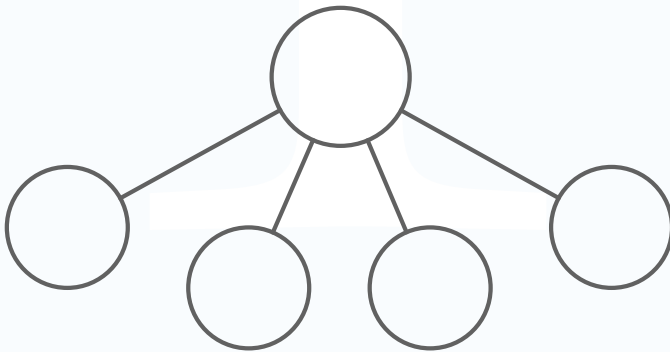
Student View: Slide 8

*Concept Development**Number Bonds*

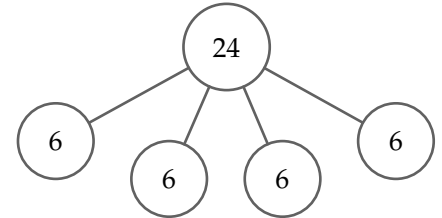
Fill in the number bond diagram that goes with this array.



- Four six's equal twenty four.
- $6 + 6 + 6 + 6 = 24$
- $4 \times 6 = 24$
- 4 and 6 are factors of 24.

*Concept Development**Number Bonds*

Solution:



Student View: Slide

Problem Set

Now you will work problems 1-6 of the lesson problem set.
(attached)

1

Problem Set

Name _____

Date _____

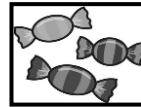
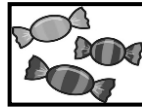
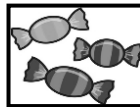
Solve Problems 1–4 using the pictures provided for each problem.

1. There are 5 flowers in each bunch. How many flowers are in 4 bunches?



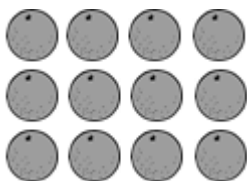
- a. Number of groups: _____ Size of each group: _____
- b. $4 \times 5 =$ _____
- c. There are _____ flowers altogether.

2. There are _____ candies in each box. How many candies are in 6 boxes?



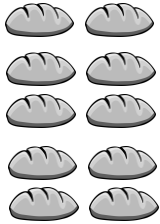
- a. Number of groups: _____ Size of each group: _____
- b. $6 \times$ _____ $=$ _____
- c. There are _____ candies altogether.

3. There are 4 oranges in each row. How many oranges are there in _____ rows?



- a. Number of rows: _____ Size of each row: _____
- b. _____ $\times 4 =$ _____
- c. There are _____ oranges altogether.

4. There are _____ loaves of bread in each row. How many loaves of bread are there in 5 rows?



a. Number of rows: _____ Size of each row: _____

b. _____ $\times$ _____ = _____

c. There are _____ loaves of bread altogether.

5. a. Write a multiplication equation for the array shown below.

X X X
X X X
X X X
X X X

b. Draw a number bond for the array where each part represents the amount in one row.

6. Draw an array using factors 2 and 3. Then, show a number bond where each part represents the amount in one row.